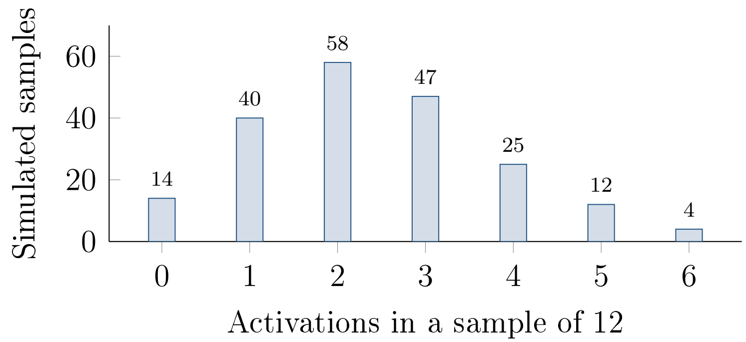
Six Activations in Twelve Users

Unit 3 · Topic 3.3 · TODAY'S PROBLEM

Suppose a developer's model says $p_0 = 0.20$ of new app users activate voice commands. An SRS of 12 users includes 6 activations, so $\hat{p} = 0.50$. The chart shows 200 simulated samples of 12 independent users under this model.

Mara: "A normal model for $\hat{p}$ gives $P(\hat{p} \geq 0.50) \approx 0.005$, so that is the chance probability I would report."

Jules: "The simulated distribution is not very normal. I would estimate the chance probability directly from its tail."



FIRST TAKE (5 min, on your sheet)

Whose method would you trust more, Mara's smooth curve or Jules's repeated trials? Give one reason from the picture.

DOK 1 **(a)** State the most common number of activations in the simulated samples and describe the distribution as roughly symmetric, skewed left, or skewed right.

DOK 2 **(b)** Use the chart to estimate $P(\hat{p} \geq 0.50)$ under $p_0 = 0.20$. Then calculate np_0 and $n(1 - p_0)$ and assess the expected-count condition for a normal approximation.

DOK 3 **(c)** In 3–4 sentences, choose a method using the simulated tail and either the shape or expected counts. Apply the 5% evidence rule. Explain how reporting 0.005 instead of 0.020 would change a reader's impression.

Optional review: follow-along (scan the code)

Work (a)–(c) from the rules on your sheet. Turn in your sheet.